

Homework 6

Alex Ojemann

2024-11-08

Problem 1

Part a

##	Stratified by Group
##	C
## n	406
## Age (years) (mean (SD))	25.91 (5.52)
## Race: Black = Yes (%)	180 (44.3)
## Race: White = Yes (%)	118 (29.1)
## Race: Native American = Yes (%)	117 (28.8)
## Race: Asian = Yes (%)	4 (1.0)
## Ethnicity: Hispanic = Yes (%)	178 (52.8)
## Years of Education (%)	
## 8-12 yrs	238 (58.6)
## LT 8 yrs	76 (18.7)
## MT 12 yrs	92 (22.7)
## Received Public Assistance = Yes (%)	309 (76.1)
## History of Hypertension = Y (%)	9 (2.2)
## History of Diabetes = Yes (%)	8 (2.0)
## Baseline Diabetes Type (%)	
## Type I	1 (0.2)
## Type II	7 (1.7)
## NA	398 (98.0)
## Body Mass Index (BMI) (mean (SD))	27.49 (6.89)
## Uses Tobacco = Yes (%)	44 (11.1)
## Baseline Cigarettes per Day (mean (SD))	0.82 (3.09)
## Uses Alcohol = Yes (%)	8 (2.0)
## Baseline Drinks per Day (mean (SD))	0.04 (0.40)
## Drug Addiction History = Yes (%)	0 (0.0)
## Previous Pregnancies = Yes (%)	303 (74.6)
## Number of Previous Pregnancies (mean (SD))	1.87 (1.82)
## Live Preterm Birth = Yes (%)	43 (10.6)
## Any Stillbirth = Yes (%)	6 (1.5)
## Spontaneous Abortion = Yes (%)	94 (23.2)
## Induced Abortion = Yes (%)	67 (16.5)
## Any Live Preterm Birth or Stillbirth = Yes (%)	169 (41.6)
## Number of Living Children (mean (SD))	1.87 (1.82)
## Number of Qualifying Teeth (mean (SD))	14.33 (6.67)
## Baseline Gingival Index (mean (SD))	1.42 (0.39)
## Baseline Bleeding on Probing (mean (SD))	69.12 (17.04)
## Baseline Pocket Depth (Avg) (mean (SD))	2.84 (0.53)
## Baseline Sites with Pocket Depth >4mm (mean (SD))	24.80 (15.94)

##	Baseline Sites with Pocket Depth >5mm (mean (SD))	9.95 (13.41)	
##	Baseline Clinical Attachment Level (Avg) (mean (SD))	1.37 (0.66)	
##	Baseline Sites with CAL >2mm (mean (SD))	41.29 (25.27)	
##	Baseline Sites with CAL >3mm (mean (SD))	14.03 (16.05)	
##	Baseline Calculus Index (mean (SD))	1.13 (0.62)	
##	Baseline Plaque Index (mean (SD))	1.23 (0.48)	
##	Gestational Age at Outcome (mean (SD))	269.18 (26.50)	
##	Birthweight (g) (mean (SD))	3180.82 (727.49)	
##	Race/Ethnicity (%)		
##	Black	127 (37.7)	
##	Hispanic (any race)	178 (52.8)	
##	Other	32 (9.5)	
##	Bleeding on Probing at Visit 5 (mean (SD))	67.35 (21.09)	
##		Stratified by Group	
##		T	SMD
##	n	184	
##	Age (years) (mean (SD))	25.99 (5.60)	0.015
##	Race: Black = Yes (%)	82 (44.6)	0.005
##	Race: White = Yes (%)	27 (14.7)	0.354
##	Race: Native American = Yes (%)	80 (43.5)	0.309
##	Race: Asian = Yes (%)	0 (0.0)	0.141
##	Ethnicity: Hispanic = Yes (%)	85 (46.2)	0.133
##	Years of Education (%)		0.032
##	8-12 yrs	107 (58.2)	
##	LT 8 yrs	33 (17.9)	
##	MT 12 yrs	44 (23.9)	
##	Received Public Assistance = Yes (%)	136 (73.9)	0.051
##	History of Hypertension = Y (%)	6 (3.3)	0.064
##	History of Diabetes = Yes (%)	9 (4.9)	0.161
##	Baseline Diabetes Type (%)		0.191
##	Type I	4 (2.2)	
##	Type II	5 (2.7)	
##	NA	175 (95.1)	
##	Body Mass Index (BMI) (mean (SD))	28.04 (7.95)	0.074
##	Uses Tobacco = Yes (%)	17 (9.3)	0.058
##	Baseline Cigarettes per Day (mean (SD))	0.90 (3.43)	0.025
##	Uses Alcohol = Yes (%)	4 (2.2)	0.013
##	Baseline Drinks per Day (mean (SD))	0.03 (0.38)	0.013
##	Drug Addiction History = Yes (%)	0 (0.0)	<0.001
##	Previous Pregnancies = Yes (%)	133 (72.3)	0.053
##	Number of Previous Pregnancies (mean (SD))	1.67 (1.73)	0.114
##	Live Preterm Birth = Yes (%)	9 (4.9)	0.215
##	Any Stillbirth = Yes (%)	2 (1.1)	0.035
##	Spontaneous Abortion = Yes (%)	52 (28.3)	0.117
##	Induced Abortion = Yes (%)	8 (4.3)	0.406
##	Any Live Preterm Birth or Stillbirth = Yes (%)	62 (33.7)	0.164
##	Number of Living Children (mean (SD))	1.67 (1.73)	0.114
##	Number of Qualifying Teeth (mean (SD))	14.41 (6.92)	0.011
##	Baseline Gingival Index (mean (SD))	1.49 (0.42)	0.173
##	Baseline Bleeding on Probing (mean (SD))	70.08 (16.99)	0.057
##	Baseline Pocket Depth (Avg) (mean (SD))	2.87 (0.59)	0.064
##	Baseline Sites with Pocket Depth >4mm (mean (SD))	25.67 (15.48)	0.055
##	Baseline Sites with Pocket Depth >5mm (mean (SD))	10.09 (13.98)	0.010
##	Baseline Clinical Attachment Level (Avg) (mean (SD))	1.50 (0.78)	0.176

```
## Baseline Sites with CAL >2mm (mean (SD))          46.10 (27.47)    0.183
## Baseline Sites with CAL >3mm (mean (SD))          16.91 (17.80)    0.170
## Baseline Calculus Index (mean (SD))                1.17 (0.61)     0.066
## Baseline Plaque Index (mean (SD))                 1.28 (0.49)     0.111
## Gestational Age at Outcome (mean (SD))            272.29 (17.48)   0.139
## Birthweight (g) (mean (SD))                      3259.16 (574.17) 0.120
## Race/Ethnicity (%)                                0.137
##   Black                                           81 (44.0)
##   Hispanic (any race)                           85 (46.2)
##   Other                                           18 (9.8)
## Bleeding on Probing at Visit 5 (mean (SD))        44.58 (21.08)    1.080
```

Part b

```
## Estimated ATE for Birthweight:
## - Estimate: -16.12937
## - Standard Error: 360.8862
## - 95% Confidence Interval: ( -723.4664 , 691.2077 )
## Estimated ATE for Proportion of Pregnancies Before 37 Weeks:
## - Estimate: -0.1536139
## - Standard Error: 0.3027535
## - 95% Confidence Interval: ( -0.7470108 , 0.4397829 )
```

Part c

```
##
## Call:
## lm(formula = Change_in_BOP ~ Group + ., data = mediator_data)
##
## Residuals:
##   Min       1Q   Median       3Q      Max
## -37.82 -11.28  -1.80   10.03   52.82
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   -17.99858    3.83427  -4.694 3.36e-06 ***
## GroupT         21.27651    1.42620  14.918 < 2e-16 ***
## Race_ethnicityHispanic (any race)  3.96835    1.54664   2.566 0.010550 *
## Race_ethnicityOther    0.20089    2.62448   0.077 0.939013
## Public.AsstceYes    -0.78964    1.57913  -0.500 0.617236
## Use.TobYes         4.87869    2.23281   2.185 0.029297 *
## N.prev.preg        0.18787    0.38793   0.484 0.628363
## Live.PTBYes        0.60978    2.40311   0.254 0.799784
## BL.GE            -9.49581    2.58543  -3.673 0.000263 ***
## BL..BOP          0.46843    0.05793   8.086 3.76e-15 ***
## BL..PD.4        -0.08197    0.06828  -1.201 0.230435
## BL..CAL.3         0.07100    0.06087   1.166 0.243963
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 15.62 on 567 degrees of freedom
```

```
## (11 observations deleted due to missingness)
## Multiple R-squared: 0.3535, Adjusted R-squared: 0.341
## F-statistic: 28.18 on 11 and 567 DF, p-value: < 2.2e-16
```

Part d

```
##
## Call:
## lm(formula = mediation_formula, data = data_imputed)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -3005.01  -272.96    69.52   372.91  1979.67
##
## Coefficients:
##                                Estimate Std. Error t value Pr(>|t|)
## (Intercept)                   3255.859     164.859  19.749 < 2e-16 ***
## GroupT                        -46.798       89.102  -0.525 0.599635
## Change_in_BOP                  -2.760       2.239  -1.233 0.218138
## Race_ethnicityHispanic (any race) -25.513      65.936  -0.387 0.698948
## Race_ethnicityOther             177.070     110.494   1.603 0.109598
## Public.AsstceYes              -117.061      66.520  -1.760 0.078983 .
## Use.TobYes                    -125.091      94.678  -1.321 0.186962
## N.prev.preg                     40.915      16.360   2.501 0.012668 *
## Live.PTBYes                   -359.095     101.248  -3.547 0.000422 ***
## BL.GE                          156.793     110.407   1.420 0.156117
## BL..BOP                        -4.774       2.584  -1.848 0.065147 .
## BL..PD.4                       8.487       2.916   2.911 0.003749 **
## BL..CAL.3                      -7.078       2.602  -2.720 0.006733 **
## GroupT:Change_in_BOP           6.980       3.475   2.009 0.045065 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 657.4 on 565 degrees of freedom
## (11 observations deleted due to missingness)
## Multiple R-squared: 0.07188, Adjusted R-squared: 0.05052
## F-statistic: 3.366 on 13 and 565 DF, p-value: 5.321e-05
```

Part e

```
## Controlled Direct Effect (CDE):
## Estimate: 15.53018
## 95% Confidence Interval: ( -63.68684 , 96.23102 )
## Natural Indirect Effect (NIE):
## Estimate: 51.87118
## 95% Confidence Interval: ( -119.3826 , 205.1225 )
```

Problem 2

Part a

```
## Proportion of patients who would benefit from treatment: 0.6007576
```

The following function will output the optimal treatment decision for new data using the random forest model (model_rf) that was trained.

```
optimal_treatment_rule <- function(new_patient_data) {  
  # Predict outcome if treated  
  treated_pred <- predict(model_rf, newdata = new_patient_data %>%  
    mutate(!!sym(treatment_var) := factor("1", levels = levels(tico_data[[treatment_var]])))  
  
  # Predict outcome if not treated  
  control_pred <- predict(model_rf, newdata = new_patient_data %>%  
    mutate(!!sym(treatment_var) := factor("0", levels = levels(tico_data[[treatment_var]])))  
  
  # Return "1" if treatment prediction is greater, otherwise "0"  
  return(ifelse(treated_pred > control_pred, "1", "0"))  
}
```

Part b

Proportion of patients who would benefit from treatment: 0.7150146

Threshold value for baseline oxygen support (ordpulmb1) for beneficial treatment: 2